Atharva Marathe

apmarathe10@gmail.com | Portfolio | +91 8329632714

Date of Birth : 12.09.2003

Experience

Tata Consultancy Services (TCS) Nov 2025 – Present
*as **Assistant Systems Engineer*** *Mumbai, India*

- **Domain: Banking and Finance**
- Support live production systems on the **TCS BaNCS** platform, handling overnight processing and interest calculation for a multinational banking client
- Work across the TCS BaNCS production environment, including **Linux-based** servers and databases, to troubleshoot issues and support critical banking processes
- Use **SQL** for data validation, investigation, and troubleshooting of production issues across banking applications and databases

Daira Edtech Jun 2025 – Oct 2025
*as **Full Stack Development and AI Integration Intern*** *Pune, India*

- Built **TensorFlow CNN** for handwriting-based learning difficulty and psychology analysis
- Applied contrast enhancement and alphabet bounding boxes for **feature extraction**
- Deployed via TFLite and TF.js (React) for cross-platform use

Education

BTech in Computer Science Engineering 2021 – 2025
Vellore Institute of Technology *Chennai, India*

- CGPA: 8.15

Higher Secondary Education (12th Class) 2021
Maharashtra State Board *Pune, India*

- Percentage : 88.00 %

Publication

CUDA-Accelerated Deep Learning Model for Alzheimer's Disease Detection
https://ieeexplore.ieee.org/document/10823113

- Published research on Alzheimer's Disease Classification using hybrid DenseNet201–VGG16 architecture with CUDA acceleration, achieving 98% accuracy on MRI dataset
- Optimized training and feature extraction using GPU acceleration and transfer learning, significantly reducing computation time for medical image analysis
- Published in IEE XPLORE 2024 International Conference

Projects

Quantum-Enhanced Generative AI for Underwater Image Enhancement

- Developed a hybrid Quantum GAN combining quantum circuits and CNNs for underwater image enhancement.
- Integrated Cirq,SymPy, and PyTorch to build trainable quantum layers within an encoder-decoder generator.
- Implemented adversarial training using PyTorch Lightning, with MSE, L1, and VGG-based perceptual losses.

Library Management System using Spring Boot

- Used Spring MVC Architecture
- Hosted on Eureka server for registering all the microservices and used Feign Client for communication between services
- Implemented UI using React.js

AWS Cloud-Based Tic Tac Toe Application

- Built React application implementing game logic, component-based architecture, and state management for a dynamic Tic Tac Toe game
- Deployed and configured the frontend using AWS S3 static website hosting
- Building backend expansion using AWS Lambda (serverless) and DynamoDB for multiplayer game state synchronization

Secure Data Embedding

- Presented at SPRINGER Conference
- Developed a data embedding system using Center-Embedded Pixel Positioning (CEPP), integrating Caesar Cipher, AES encryption, and multi-layer steganography to enhance data security and image quality
- Validated the approach using metrics such as PSNR, SSIM, and entropy, ensuring high visual fidelity and secure communication

Certifications

- **AWS Certified Solutions Architect – Associate**
  Amazon Web Services
- **Google Cloud Computing Foundation**
  Google Cloud
- **Web-Verse: Web Development and Design**
  Android Club, VIT Chennai
- **Ethical Hacking**
  NPTEL

Skills and Interests

- AWS Cloud Services
- Backend (Spring, Spring Boot, Microservices, REST API, Feign Client)
- Website Development Frontend (HTML, CSS, React)
- Machine Learning Frameworks: PyTorch (with CUDA optimization)
- Video Editing (Adobe Premiere Pro, Adobe After Effects)
- Languages – English, German, Hindi, Marathi
- Hobbies – Playing badminton, chess and listening to music